The Real Effects of Funding for Lending:
Evidence from the European Central Bank's Targeted Longer-Term Refinancing Operations*

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Abstract

I study the real effects of funding-for-lending programmes using the European Central Bank's Targeted Longer Term Refinancing Operations. Combining bank balance sheets with firm-level financial data and firm–bank links, I construct a matched dataset covering 470,000 firms across 14 euro-area countries. I find that banks increasing their central bank borrowing expand lending, while firms connected to these banks experience higher debt and deposit growth. However, the additional financing does not translate into higher investment. These findings reveal a distinction between the financial and real effects of targeted central bank funding, highlighting the limits of credit expansion in stimulating investment.

1 Introduction

Funding-for-lending programmes have become an increasingly important instrument of monetary policy over the past decade, and their scale has expanded substantially. For example, in the United Kingdom, the Bank of England provided £127 billion through its Term Funding Scheme between 2016 and 2018, while in the euro area, outstanding borrowing under the European Central Bank's Targeted Longer-Term Refinancing Operations (TLTRO III) reached €2.2 trillion in 2021. By providing banks with long-term funding at favourable conditions linked to their lending performance, these programmes aim to stimulate credit supply and support real economic activity.

Yet, despite the importance of these programmes, existing research has primarily focused on their effects on banks' funding conditions and lending behaviour. While this evidence sheds light on the transmission of central bank funding through the banking sector, it provides a more limited understanding of how firms ultimately use the additional credit. An expansion in bank lending does not necessarily translate into greater real investment, as firms may instead accumulate liquidity or refinance existing obligations. Understanding how firms respond to the additional financing generated by funding-for-lending programmes is therefore essential to assessing their effectiveness.

In this paper, I study the real effects of funding-for-lending programmes by examining how central bank funding is transmitted through banks to firms and whether it ultimately translates into greater investment. To this end, I exploit the European Central Bank's (ECB) third series of Targeted Longer-Term Refinancing Operations (TLTRO III), focusing particularly on its pandemic recalibration. Beginning in March 2020, the ECB substantially increased banks' borrowing allowances and introduced exceptionally favourable funding conditions, allowing qualifying banks to borrow at interest rates as low as -1 percent. These measures triggered a sharp expansion in central bank funding, while banks differed considerably in the extent to which they relied on the programme. This variation provides a setting for investigating the transmission of targeted central bank funding to firms' financial and investment decisions.

To investigate this question, I construct a matched firm–bank dataset combining firm-level financial statements and firm–bank links from Bureau van Dijk's Orbis database with consolidated banking-group balance sheets from the European Banking Authority (EBA)

Transparency Exercise. The resulting matched dataset contains approximately 470,000 firms and 3.52 million firm-year observations across 14 euro-area countries over the period 2015–2023. Since bank-level TLTRO III borrowing is not publicly disclosed, I use changes in banks' borrowed reserves as a proxy for changes in their TLTRO borrowing. This measure is supported by the close correspondence between aggregate borrowed reserves and outstanding TLTRO borrowing, which exhibit a correlation of 0.99 over their common sample period. By combining these data sources, I can examine how changes in central bank funding are associated with both banks' lending decisions and the financial and real outcomes of their linked firms.

My empirical analysis proceeds in three steps. First, using quarterly bank-level data, I estimate two-way fixed effects regressions to investigate the relationship between changes in borrowed reserves and bank lending to non-financial corporations and households. Second, I use the matched firm–bank dataset to examine whether firms connected to banks increasing their central bank borrowing experience larger changes in debt, deposits, and investment. Finally, I exploit the pandemic recalibration of TLTRO III in a continuous-treatment difference-in-differences event-study design. I construct firm-specific exposure to the intervention using the increase in borrowed reserves of linked banks between the end of 2019 and the end of 2020. Comparing firms with different levels of exposure before and after the intervention allows me to trace their financial and investment responses over time. The empirical specifications include unit fixed effects and time fixed effects, with country-by-time fixed effects in my preferred specifications.

The results reveal a clear distinction between the financial and real effects of central bank funding. At the bank level, I find that increases in borrowed reserves are positively and significantly associated with lending, particularly to non-financial corporations. At the firm level, firms connected to banks increasing their central bank borrowing experience significantly higher debt and deposit growth. However, I find no corresponding increase in investment. These findings remain robust to alternative fixed effects specifications and to constructing firm-level exposure using changes in banks' corporate lending rather than changes in their central bank borrowing.

The event-study analysis provides further evidence of this distinction. Firms more exposed to the pandemic recalibration of TLTRO III experience significantly higher debt

and deposit growth in 2020, while the corresponding investment response is small and statistically insignificant. In subsequent years, debt growth differences largely disappear, while deposit growth becomes significantly negative in 2023. Investment does not exhibit a statistically significant positive response in any post-treatment year, and the estimated coefficient for 2023 is negative and weakly significant. These results suggest that, although targeted central bank funding is associated with an expansion in firms' borrowing and liquidity holdings, it does not necessarily translate into greater real investment.

This paper contributes to the literature on funding-for-lending programmes and the transmission of unconventional monetary policy through the banking sector. Existing studies have primarily examined how targeted central bank funding affects banks' lending behaviour and credit supply, while evidence on its ultimate real effects remains limited and mixed. By combining bank balance sheets with firm-level financial information and observed firm–bank links, I trace the transmission of central bank funding beyond the banking sector. Importantly, the joint analysis of debt, deposits, and investment allows me to distinguish the financial responses of firms from their real investment decisions. The paper thus provides evidence that the effectiveness of funding-for-lending programmes in expanding credit need not imply a corresponding increase in productive capital accumulation.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the institutional background of funding-for-lending programmes and the ECB's TLTROs. Section 4 presents the data and the construction of the matched firm–bank dataset. Sections 5 and 6 report the bank-level and firm-level analyses, respectively. Section 7 presents the event-study analysis of the pandemic recalibration of TLTRO III.

2 Related Literature

This paper is linked to three strands of literature on targeted central bank refinancing, central bank liquidity provision, and the real effects of financial constraints.

The first strand examines the effects of funding-for-lending programmes on bank lending and economic activity. Benetton and Fantino (2021) study the ECB's first series of Targeted Longer-Term Refinancing Operations (TLTROs), exploiting the programme's bor-

rowing allowance rules and transaction-level data from the Italian credit register. They find that targeted central bank funding increases lending volumes and reduces interest rates, with stronger effects in more competitive banking markets and among smaller and safer firms. Similarly, Esposito et al. (2020) investigate TLTRO II and document an expansion in lending to Italian firms and a reduction in borrowing costs. Andreeva and García-Posada (2021) study the direct and indirect effects of TLTRO participation on euro-area banks' lending policies, showing that the programme affects credit conditions both through participating institutions and through competitive responses by other banks. Together, these studies establish that targeted central bank funding can ease credit conditions, but concentrate primarily on banks' lending decisions rather than firms' subsequent financial and real adjustments.

More recent research examines the extraordinary refinancing measures introduced during the COVID-19 pandemic. Altavilla et al. (2023) investigate the interaction between monetary and prudential policy interventions and show how these measures supported banks' lending capacity during the crisis. Corte Coi et al. (2025) study the pandemic recalibration of TLTRO III and its effects on banks' loan portfolios, finding evidence that the favourable funding conditions also encouraged mortgage lending, despite mortgages being excluded from the programme's eligible lending categories. This evidence suggests that the effects of targeted refinancing may extend beyond the intended credit segments, making it important to understand how additional funding is allocated and ultimately used.

A smaller literature investigates the real effects of targeted refinancing programmes. Perdichizzi et al. (2023) examine the consequences of TLTROs for local economic development across Italian provinces. They find no beneficial effects on local economic activity and document a decline in average firm investment, particularly among small and micro firms. In contrast, Laine and Nelimarkka (2023) investigate the aggregate effects of the ECB's TLTROs between 2014 and 2021 and find positive responses in bank lending, GDP, and investment. Outside the euro area, Churm et al. (2021) examine the macroeconomic consequences of the Bank of England's Funding for Lending Scheme, finding positive effects on output and inflation. These studies provide mixed evidence on whether the improvements in bank funding conditions associated with targeted refinancing programmes translate into stronger real economic activity. My paper contributes to

this literature by examining the pandemic expansion of TLTRO III through firm-specific exposure constructed from observed firm–bank links, allowing me to trace the dynamic responses of firms' borrowing, liquidity holdings, and investment.

The second strand of literature investigates the transmission of central bank liquidity provision through the banking sector. An important question concerns how banks allocate the additional funding obtained from central banks. Drechsler et al. (2016) study lender-of-last-resort borrowing during the European sovereign debt crisis, documenting that less capitalized banks borrow more from the central bank and pledge riskier collateral. Crosignani et al. (2020) examine the ECB's three-year Longer-Term Refinancing Operations (LTROs) and show that Portuguese banks used the additional funding to purchase domestic government bonds. Similarly, Carpinelli and Crosignani (2021) study the Italian banking sector and find that longer-maturity central bank funding supported banks' lending capacity, while also being used to purchase government securities and replace maturing wholesale funding. These findings highlight that liquidity provision can affect banks' funding structures and asset allocation, without necessarily translating entirely into additional credit to firms.

A related question is whether central bank liquidity provision stimulates bank lending and, ultimately, real economic activity. Andrade et al. (2019) exploit the ECB's three-year LTROs and matched bank–firm credit data to document positive effects on lending to French firms. Berger et al. (2017) examine the Federal Reserve's emergency liquidity facilities during the Global Financial Crisis and find that recipient banks increased lending relative to comparable institutions. In contrast, Wu (2015) studies government capital support and Federal Reserve liquidity programmes using syndicated loan data and finds limited effects on credit supply in this market. Moving beyond bank lending, Rüden et al. (2023) investigate the effects of the ECB's three-year LTROs on corporate investment. They find that firms connected to banks receiving additional central bank funding experienced weaker investment outcomes, particularly when their banks were financially distressed. Taken together, these studies show that the transmission of central bank liquidity to bank credit supply is heterogeneous and that improvements in banks' funding conditions do not necessarily lead to greater real investment.

The third strand of literature examines how financial constraints and changes in credit

supply affect firms' real decisions. A seminal contribution is Chodorow-Reich (2014), who exploits differences in firms' exposure to financially distressed banks during the Global Financial Crisis and finds substantial negative effects on employment, particularly among smaller firms. Similarly, Cingano et al. (2016) exploit the disruption of interbank funding markets during the crisis and document significant effects of bank credit supply contractions on firms' investment, employment, and production. These studies establish that disruptions to bank financing can have important consequences for real economic activity.

More closely related to my analysis, García-Posada Gómez (2019) investigates the effects of credit constraints on firm investment and employment using survey data from twelve European countries. He exploits the allocation rules of the ECB's TLTROs to construct a country-level instrument for firms' credit constraints and finds that these constraints significantly reduce investment in fixed assets, while having no statistically significant effects on employment or inventories. Unlike García-Posada Gómez (2019), whose identification strategy relies on country-level variation, I construct firm-specific exposure using observed firm–bank links, allowing exposure to vary across firms within the same country.

My paper complements this literature by examining an expansionary credit intervention rather than a contraction in bank funding. In addition, by jointly studying firms' debt, deposits, and investment, I investigate not only whether additional financing is associated with greater investment, but also the timing of firms' financial adjustments following the policy intervention.

3 Funding for Lending Programmes

3.1 Funding for Lending Programmes Across Central Banks

Following the Global Financial Crisis, major central banks increasingly complemented conventional monetary policy with funding programmes aimed at supporting the supply of bank credit. These programmes provide banks with long-term funding under preferential conditions, linking the cost or availability of central bank funding to the banks' lending behaviour. Their common objective is to strengthen the bank lending channel of monetary policy by encouraging banks to expand credit to the real economy.

Although their design differs across jurisdictions, these programmes share the objective

of improving banks' funding conditions and strengthening monetary policy transmission. The Bank of England pioneered the use of targeted funding programmes with the introduction of the Funding for Lending Scheme (FLS) in 2012, later followed by the Term Funding Scheme (TFS) and the Term Funding Scheme with additional incentives for SMEs (TFSME). The European Central Bank introduced the Targeted Longer-Term Refinancing Operations (TLTROs) in 2014 and subsequently expanded the programme through additional rounds. Similarly, the Bank of Japan and the People's Bank of China implemented a variety of funding and relending programmes aimed at encouraging bank lending and supporting economic activity.

The Federal Reserve adopted a markedly different approach. During both the Global Financial Crisis and the COVID-19 pandemic, it introduced numerous liquidity and credit facilities to stabilize financial markets and support the flow of credit. However, unlike the ECB and several other major central banks, the Federal Reserve did not implement a long-term targeted refinancing programme providing banks with funding at preferential rates explicitly linked to their lending activity. As a result, it did not create an institutional setting directly comparable to the ECB's TLTROs for studying the effects of targeted central bank funding.

Among these interventions, TLTRO III stands out for both its scale and the exceptionally favourable borrowing conditions introduced during the COVID-19 pandemic. The programme combined very large take-up by euro-area banks with borrowing rates that, under certain conditions, reached as low as -1 percent, creating unusually strong incentives for banks to obtain central bank funding. These features make TLTRO III a particularly attractive setting for studying the real effects of targeted central bank funding.

3.2 The ECB's Targeted Longer-Term Refinancing Operations

In June 2014, in the aftermath of the euro area sovereign debt crisis, financial fragmentation and ongoing bank deleveraging continued to impair the transmission of monetary policy through the bank lending channel. In this context, the ECB announced the introduction of targeted longer-term refinancing operations (TLTROs) to stimulate bank lending. The ECB provided banks with long-term central bank funding on favourable terms linked to their lending performance. In particular, access to more abundant and

favourable funding conditions was contingent on banks increasing their eligible lending relative to a bank-specific lending benchmark.

The first series of TLTROs, TLTRO I, consisted of eight operations conducted between September 2014 and June 2016, all maturing in September 2018. Banks were initially entitled to borrow up to 7 percent of their stock of eligible loans as of April 2014[1]. From March 2015 onward, they could obtain additional funding based on their eligible net lending relative to a bank-specific benchmark[2]. The additional borrowing allowance was equal to up to three times the amount of eligible net lending in excess of the benchmark, creating a marginal incentive to expand lending during the programme. The interest rate was fixed over the life of each operation at the main refinancing operations (MRO) rate prevailing at take-up, plus 10 basis points. Banks that failed to meet the relevant lending benchmark were subject to early repayment of their TLTRO funding. Total allotments under TLTRO I amounted to approximately €430 billion.

In March 2016, the ECB announced a second series of targeted longer-term refinancing operations, TLTRO II, as part of a broader package aimed at reinforcing its accommodative monetary policy stance and strengthening the transmission of monetary policy. TLTRO II consisted of four operations conducted between June 2016 and March 2017, each with a maturity of four years. Banks could borrow up to 30 percent of their stock of eligible loans as of January 2016, less any amounts still outstanding from the first two TLTRO I operations conducted in 2014. Unlike under TLTRO I, the borrowing allowance was determined ex ante and lending performance no longer affected the quantity of funding available at the margin. Instead, lending performance affected the price of funding: the interest rate applied to TLTRO II was equal to the MRO rate, but banks whose net lending exceeded their bank-specific benchmark could obtain a lower interest rate, down to the deposit facility rate (DFR) prevailing at allotment. The rate reduction increased with lending performance, with banks receiving the maximum reduction once their eligible lending exceeded the benchmark by 2.5 percent. Total allotments under TLTRO II

[1]Eligible loans comprise loans to euro-area non-financial corporations and households, excluding loans to households for house purchase.

[2]The lending benchmark was based on each bank's eligible net lending over a twelve-month reference period prior to the announcement of TLTROs. For banks with positive net lending over this period, the benchmark was set to zero, implying that these banks had to maintain at least their existing level of lending. For banks with negative net lending, it reflected the bank's past lending trajectory, implying that these banks could continue their lending contraction but not increase the speed thereof.

amounted to approximately €740 billion.

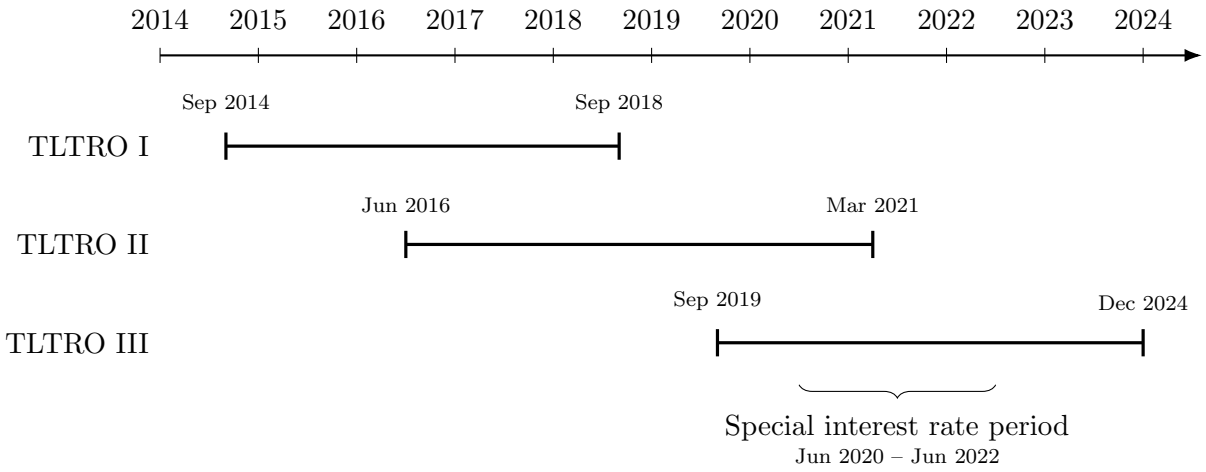
In March 2019, the ECB announced a third series of targeted longer-term refinancing operations, TLTRO III, against a backdrop of moderating euro area growth and continued uncertainty. The programme consisted of quarterly operations between September 2019 and March 2021, with a maturity of three years, a borrowing allowance of up to 30 percent of eligible loans outstanding as of February 2019, and a pricing scheme very similar to that of TLTRO II.

Beginning in March 2020, the ECB progressively recalibrated the terms of TLTRO III in response to the COVID-19 pandemic. The maximum borrowing allowance was increased from 30 to 50, and subsequently to 55, percent of eligible loans outstanding as of February 2019, the lending performance threshold was reduced from 2.5 percent to zero, three further TLTRO III operations were added, and the per-operation borrowing limit was removed. As part of the recalibration, the ECB introduced a special interest rate period, running from June 2020 to June 2022[3], during which the interest rate applied to all outstanding TLTRO III operations was reduced substantially. For counterparties meeting the lending performance threshold, the rate could be as low as 50 basis points below the average deposit facility rate over the period, subject to a floor of -1 percent. With the deposit facility rate at -0.50 percent over this period, eligible banks could therefore obtain TLTRO funding at rates as low as -1 percent and generate a gross interest margin of 50 basis points by placing the funds at the ECB's deposit facility. Total allotments under TLTRO III amounted to approximately €2.34 trillion — more than three times the amount allotted under TLTRO II and more than five times the amount allotted under TLTRO I.

Figure 1 provides a timeline of the three TLTRO programmes and identifies the special interest rate period, while Figure 2 shows the evolution of outstanding TLTRO borrowing over time. The scale of the TLTRO programmes increased markedly over time, with outstanding amounts expanding particularly sharply at the beginning of the special interest rate period.

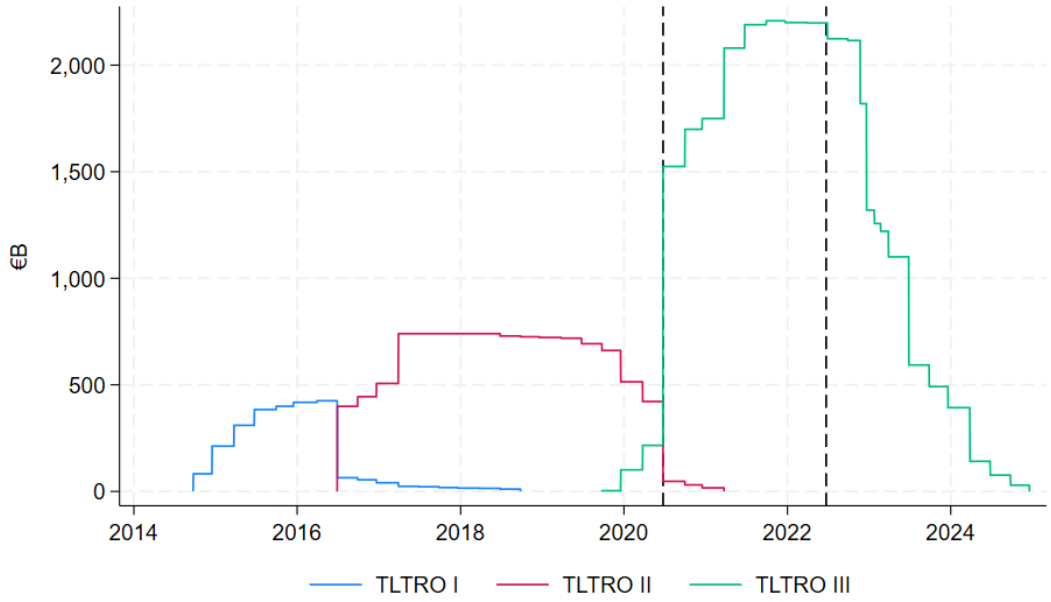
[3]The ECB formally distinguishes between the Special Interest Rate Period (SIRP), running from June 2020 to June 2021, and the Additional Special Interest Rate Period (ASIRP), running from June 2021 to June 2022. Throughout the paper, we refer to the two periods jointly as the *special interest rate period*, as they share the same policy rationale.

Figure 1: Timeline of the ECB's Targeted Longer-Term Refinancing Operations



Notes: The figure reports the time span of TLTRO I, TLTRO II, and TLTRO III, measured from the first allotment to the maturity of the last operation. The brace denotes the special interest rate period introduced under the pandemic recalibration of TLTRO III.

Figure 2: Outstanding TLTRO Borrowing



Notes: Outstanding TLTRO borrowing is constructed from ECB data on the allotments, early repayments, and maturities of individual TLTRO operations. Allotments and maturities are obtained from the open market operations' history database, while information on voluntary early repayments is extracted from the open market operations ad-hoc communications section on the ECB's web portal. Amounts are aggregated across individual operations within each TLTRO series and reported in billions of euros.

4 Data

4.1 Firms' Balance Sheets

I obtain firm-level balance sheet data from Bureau van Dijk's Orbis database (downloaded in August 2025). The data cover the period from 2015 to 2023 and include 53.60 million firm-year observations for 9.93 million distinct firms across the 19 countries that comprised the euro area at the beginning of the sample period (EA19)[4].

The dataset provides detailed information on firms' balance sheets and income statements. The three main variables I use are cash and cash equivalents, which mostly comprise deposits; debt, which mostly consists of bank loans; and fixed assets, whose change over time I use to estimate investment.

4.2 Firm–Bank Links

I obtain information on firm–bank links from the "Bankers" folder of the Orbis database. The data cover 14 of the EA19 countries, excluding Belgium, Estonia, Finland, Italy, and Slovakia, for which the database contains no or very limited information on firm–bank links. The resulting dataset contains 4.95 million firm–bank links covering 3.95 million distinct firms, corresponding to an average of 1.25 links per firm.

The reported links correspond to the most recent firm–bank links communicated by the Orbis data provider. Historical information on firm–bank links is not available. However, given the persistence of bank–firm relationships documented in the European relationship-lending literature (Agostino and Trivieri 2018), I use the reported links as a proxy for firms' bank links over the sample period. Importantly, I observe only the existence of a firm–bank link and not its intensity; in particular, I do not observe the amount that a firm borrows from each linked bank.

Each firm–bank link in Orbis contains the firm's BvD identifier and the name of the bank, but no identification code for the bank. Bank names are not standardized across

[4] As of January 1, 2015, the euro area comprised 19 countries: Austria, Belgium, Cyprus, Estonia, Finland, France, Germany, Greece, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, the Netherlands, Portugal, Slovakia, Slovenia, and Spain. Croatia joined the euro area on January 1, 2023, and Bulgaria on January 1, 2026. I exclude Croatia to make the panel of countries balanced and Bulgaria because it entered the euro area after the end of my sample period.

observations and may contain branch-specific information, such as branch names or branch codes, within the same field. This makes matching the reported bank names to bank-level data non-trivial.

To this aim, I first standardize the reported bank names by converting them to lowercase and removing branch-specific information, including branch names and codes. I also remove additional information reported in parentheses or following a comma, eliminate punctuation, and normalize whitespace. The cleaning procedure yields 5,824 distinct bank names.

I then obtain bank entity names and their corresponding entity codes from the European Banking Authority (EBA) Credit Institutions Register. I match the standardized Orbis bank names to the EBA entity names using a fuzzy string-matching procedure based on token similarity.

The procedure decomposes each bank name into individual tokens separated by spaces. For example, "Deutsche Bank AG" is represented by the three tokens "Deutsche", "Bank", and "AG". I apply logarithmic weights to the tokens when computing the similarity score, assigning less weight to frequently occurring terms and more weight to rare terms. This reduces the contribution of generic terms that are less informative for identifying the underlying institution, such as legal-form descriptors (e.g., "AG", "SA", "Ltd", and "GmbH") and common banking terms (e.g., "bank", "raiffeisenbank", "caixa", "credito", "mutuo", and "rural"), while giving greater weight to distinctive components of bank names (e.g., "Unicredit", "BBVA", and "Commerzbank"). For each Orbis bank name, I retain the EBA entity with the highest similarity score among matches with a score of at least 0.5. I also manually validate a random sample of 200 matched bank names, of which 93.5% are correctly identified. This validation rate suggests a high degree of matching accuracy.

Of the initial 5,824 clean bank names, 3,044 can be matched to EBA entities, corresponding to 2,001 distinct entities (multiple Orbis bank names may map to the same underlying EBA entity) which belong to 1,922 banking groups. Overall, the matching procedure assigns an entity code to 3.93 million firm–bank links, corresponding to about 79.6% of the links in the original dataset.

4.3 Banks' Balance Sheets

I obtain consolidated balance sheet data for euro area banking groups from the EBA EU-wide Transparency Exercise. The Transparency Exercise is based on supervisory reporting data and provides harmonized information on European banking groups. The sample is designed to cover the largest banking groups and represents a substantial share of the EU/EEA banking sector in terms of total assets. Its composition varies across rounds, reflecting changes in the population covered by the EBA as well as developments such as mergers, restructurings, and the inclusion or exclusion of individual institutions.

Combining the different rounds from spring 2020 to 2025, I construct a quarterly panel covering the period from September 2019 to June 2025. Earlier rounds of the Transparency Exercise are also available, but do not contain information on banks' liabilities, which is essential for my analysis; I am therefore unable to extend the panel further back than September 2019. I restrict the sample to banking groups headquartered in the euro area, obtaining a dataset of 2,350 observations for 129 banking groups across all EA19 countries except Slovakia.

The Transparency Exercise contains a wide range of information on banks' balance sheets, capital positions, profitability, risk exposures, leverage, and asset quality. For my analysis, the key variable is "Deposits from central banks", which measures funding obtained by each banking group from central banks and is commonly referred to in the literature as borrowed reserves. I also obtain total assets, which I use to scale this variable when constructing my main independent variable.

4.4 Borrowed Reserves as a Proxy for TLTRO Borrowing

Bank-level TLTRO borrowing is not publicly disclosed. I therefore proxy banks' participation in the programme using their total borrowing from central banks, commonly referred to as *borrowed reserves*. In the EBA Transparency Exercise, this variable corresponds to the balance-sheet item *Deposits from central banks*, which measures the funding obtained by each banking group from central banks. Although borrowed reserves may also include funding obtained through refinancing operations other than TLTROs, I show that TLTRO borrowing accounts for the vast majority of Eurosystem refinancing during the period relevant to my analysis.

To illustrate this, Figure 3 compares aggregate outstanding TLTRO borrowing with total borrowed reserves of euro-area banks, using data from the ECB Data Portal. The two series closely track each other, particularly from 2017 onward, when TLTROs account for nearly all outstanding central bank refinancing. Both series increase sharply following the pandemic recalibration of TLTRO III in 2020, remain at historically high levels until 2022, and decline rapidly thereafter. Their close correspondence suggests that changes in borrowed reserves largely reflect changes in TLTRO borrowing during this period.

I then examine whether this close correspondence also holds for the banking groups included in my sample. Figure 4 compares aggregate borrowed reserves from the EBA Transparency Exercise with total outstanding TLTRO borrowing in the Eurosystem. However, the two series need not coincide in levels due to differences in their definitions and coverage.

On the one hand, aggregate borrowed reserves in the EBA sample may exceed total outstanding TLTRO borrowing because the data are reported at the consolidated banking-group level and may therefore include funding obtained from non-euro-area central banks by foreign subsidiaries and other entities within the group. On the other hand, aggregate borrowed reserves in my sample may fall below total outstanding TLTRO borrowing because the EBA Transparency Exercise covers only a subset of euro-area banks, primarily large banking groups, whereas the aggregate TLTRO series includes all participating counterparties.

Despite these differences, the two series exhibit remarkably similar dynamics, with a correlation of 0.9914 over their common sample period. This close correspondence supports the use of changes in borrowed reserves as a proxy for banks' TLTRO borrowing in the empirical analysis that follows.

4.5 Dataset Construction

Figure 5 summarizes the construction of the final matched firm-bank dataset, from the initial firm-level balance sheet data to the estimation sample used in the empirical analysis.

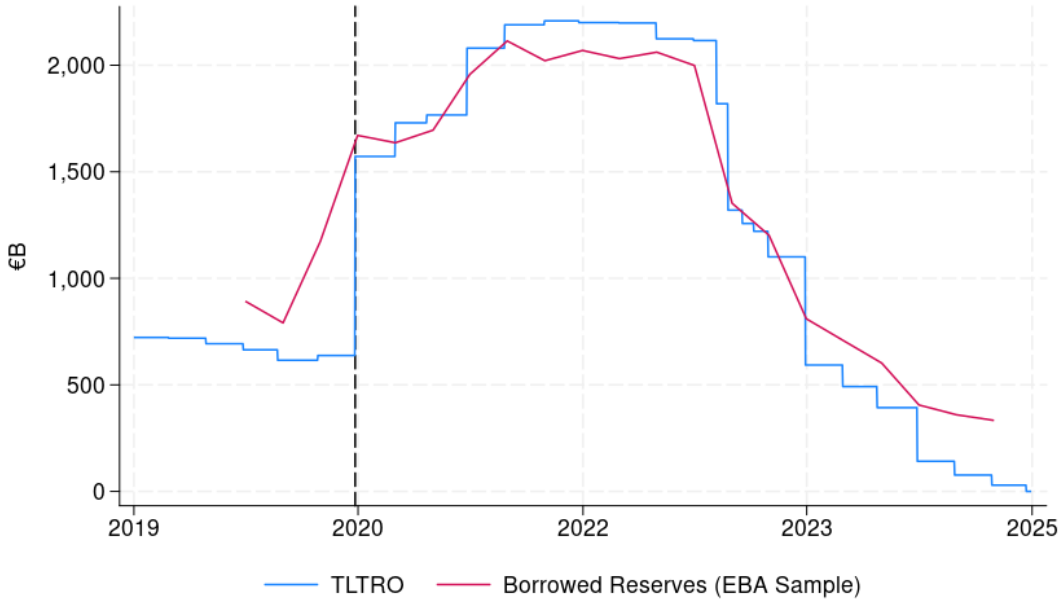
I start from the full Orbis balance sheet panel described in Section 4.1, covering 53.60 million firm-year observations for 9.93 million firms across the 19 countries of the EA19. I first merge this panel with the firm-bank links described in Section 4.2. Firm-bank

Figure 3: TLTRO Borrowing and Eurosystem Borrowed Reserves



Notes: The figure compares total outstanding TLTRO borrowing in the Eurosystem with aggregate borrowed reserves of euro-area banks. Borrowed reserves are obtained from the ECB Data Portal, while outstanding TLTRO borrowing is constructed from individual TLTRO allotments, early repayments, and maturities, as described in Figure 3. The dashed vertical line marks the beginning of the special interest rate period in June 2020. Amounts are reported in billions of euros.

Figure 4: TLTRO Borrowing and Borrowed Reserves in the EBA Sample



Notes: The figure compares total outstanding TLTRO borrowing in the Eurosystem with aggregate borrowed reserves of the euro-area banking groups included in the EBA Transparency Exercise sample. Borrowed reserves correspond to the Deposits from central banks balance-sheet item and are aggregated across banking groups in each quarter. The TLTRO series includes outstanding borrowing under all TLTRO programmes. The dashed vertical line marks the beginning of the special interest rate period in June 2020. The correlation between the two series over their common sample period is 0.9914. Amounts are reported in billions of euros.

Figure 5: Dataset Construction



Notes: The figure illustrates the construction of the matched firm-bank dataset. Each box reports the number of firm-year observations, distinct firms, and countries remaining after each step.

links information is not available or very limited for Belgium, Estonia, Finland, Italy, and Slovakia; I therefore exclude these five countries from the analysis. In addition, firm-bank links are not available for all firms for which balance-sheet information is observed. As a consequence of these two facts, this step restricts the sample substantially to 10.42 million firm-year observations for 1.47 million firms in 14 countries.

I then retain only the firm-bank links for which the reported bank name could be matched to an EBA entity code through the fuzzy string-matching procedure described in Section 4.2 and merge the resulting sample with the bank-level balance sheet data from the EBA Transparency Exercise described in Section 4.3. Because only a fraction of banks can be merged with EBA entity codes and the Transparency Exercise covers only a subset of the largest euro-area banking groups, this step results in a substantial reduction in sample size, to 3.52 million firm-year observations for 470,000 firms.

Table 1 reports descriptive statistics for the firm-level variables used in the empirical analysis.

4.6 Aggregate Data

I obtain aggregate data on euro area banks and monetary policy activity to construct some of the figures shown in this paper from the ECB Data Portal.

Table 1: Descriptive Statistics

Variable	Mean	SD	P1	P25	Median	P75	P99
Debt	7,243	259,719	0	10	147	741	68,129
Deposits	1,592	74,650	0	20	112	445	14,629
Fixed Assets	18,843	1,601,408	0	39	242	1,191	141,115
Total Assets	30,928	1,829,194	3	395	1,201	3,931	254,417
Debt Growth (% TA)	1.13	13.42	-38.00	-3.17	0.00	1.56	70.48
Deposits Growth (% TA)	2.16	14.89	-40.52	-2.81	0.14	5.34	72.83
Investment (% TA)	1.63	12.67	-33.06	-2.01	-0.09	1.83	75.83
TLTRO III Exposure	5.16	3.07	-0.01	2.65	5.28	6.67	13.43

Notes: This table reports descriptive statistics for the firm-level variables used in the empirical analysis. Debt, deposits, fixed assets, and total assets are reported in thousands of euros. Debt growth and deposits growth are measured as annual changes in the corresponding balance sheet items, normalized by lagged total assets and expressed in percentage points. Investment is measured as the annual change in fixed assets, normalized by lagged total assets and expressed in percentage points. TLTRO III exposure is a continuous treatment variable constructed as described below in Section **??**. The table reports the mean, standard deviation (SD), and selected percentiles of the distributions.

5 Bank-Level Analysis

I begin by examining whether increases in central bank funding are associated with an expansion in bank lending. Using the quarterly bank balance sheet data from the EBA Transparency Exercise covering the period from 2019Q4 to 2025Q2, I investigate the relationship between changes in borrowed reserves and changes in lending to non-financial corporations and households. This analysis represents the first step in tracing the transmission of central bank funding to the real economy, before examining how firms respond to changes in their lending banks' funding conditions.

I estimate the following two-way fixed effects specification:

$$\Delta Y(\%TA)_{i,t} = \beta \Delta BR(\%TA)_{i,t} + FE_i + FE_t + \epsilon_{i,t},$$

where $Y(\%TA)_{i,t}$ denotes either loans to non-financial corporations or consumer loans of banking group i in quarter t, and $BR(\%TA)_{i,t}$ denotes its borrowed reserves, used as a proxy for TLTRO borrowing as discussed in Section 4.4. The specification includes bank fixed effects, FE_i, which absorb time-invariant differences across banking groups, and quarter fixed effects, FE_t, which control for common macroeconomic and monetary

policy developments. Standard errors are clustered at the banking-group level.

Throughout the paper, I normalize changes in balance-sheet variables by lagged total assets, expressing them as a percentage of the size of the corresponding bank or firm. Specifically, for any balance-sheet variable X, I define:

$$\Delta X(\%TA)_{i,t} = \frac{X_{i,t} - X_{i,t-1}}{TA_{i,t-1}} \times 100,$$

where $TA_{i,t-1}$ denotes the total assets of bank or firm i in the preceding period.

This normalization allows me to compare changes in balance-sheet variables across institutions of different sizes and provides a common scale for interpreting the estimated coefficients. Using lagged rather than contemporaneous total assets also avoids introducing a mechanical relationship between changes in the variable of interest and changes in the denominator during the same period.

I estimate four alternative specifications. The first includes bank and quarter fixed effects, while the second replaces quarter fixed effects with country-by-quarter fixed effects to control for country-specific macroeconomic conditions and changes in credit demand. The third additionally includes the lagged change in borrowed reserves to account for potential delayed adjustments in bank lending. Finally, the fourth specification normalizes changes in borrowed reserves and lending by contemporaneous rather than lagged total assets.

Table 2 reports the results for loans to non-financial corporations. Column (1) shows a positive and statistically significant association between changes in borrowed reserves and bank lending. The estimated coefficient of 0.0683 indicates that a one-percentage-point increase in borrowed reserves relative to total assets is associated with an increase in loans to non-financial corporations of approximately 0.068 percentage points of total assets. The coefficient increases slightly to 0.0770 when country-by-quarter fixed effects are introduced in column (2). This is my preferred specification, as it accounts for country-specific macroeconomic conditions and credit demand shocks while exploiting variation across banks within the same country and quarter. Although statistically significant, the estimated coefficient is relatively small in economic magnitude. In particular, a one-euro increase in borrowed reserves is associated with an increase of approximately 7.7 cents in loans to non-financial corporations within the same quarter, indicating a relatively mod-

Table 2: Borrowed Reserves and NFC Loans

	(1)	(2)	(3)	(4)
Dependent variable		Δ NFC Loans (% TA)		
Δ Borr. Res. (% TA)	0.0683***	0.0770***	0.0803***	0.0691***
	(0.0220)	(0.0262)	(0.0264)	(0.0258)
Δ Borr. Res. (% TA)$_{t-1}$			-0.00253	
			(0.0186)	
Observations	1,956	1,890	1,779	1,890
R-squared	0.201	0.382	0.380	0.383
Unit FE	Bank	Bank	Bank	Bank
Time FE	Quarter	Quarter-Country	Quarter-Country	Quarter-Country
Total assets	Lagged	Lagged	Lagged	Current

Notes: Standard errors clustered at the bank level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

est contemporaneous relationship between central bank funding and corporate lending. The coefficient remains positive and statistically significant across the remaining specifications. In column (3), the coefficient on lagged changes in borrowed reserves is small and statistically insignificant, suggesting that the association is primarily contemporaneous. Finally, using contemporaneous total assets for normalization in column (4) yields a similar coefficient of 0.0691.

Table 3 reports the corresponding results for consumer loans. As in the case of lending to non-financial corporations, the estimated coefficients are positive and statistically significant across all specifications. In my preferred specification, reported in column (2), the coefficient is 0.0106, indicating that a one-euro increase in borrowed reserves is associated with an increase of approximately 1.1 cents in consumer loans within the same quarter. This coefficient is substantially smaller than that estimated for loans to non-financial corporations, which may partly reflect the smaller share of consumer loans in banks' total assets. The coefficient remains stable when including lagged changes in borrowed reserves, which are not statistically significant, and decreases slightly when using contemporaneous total assets for normalization.

Overall, the results reveal a positive and statistically significant association between changes in borrowed reserves and bank lending, particularly to non-financial corporations. However, the relatively small magnitude of the estimated coefficients suggests that

Table 3: Borrowed Reserves and Consumer Loans

	(1)	(2)	(3)	(4)
Dependent variable		Δ Consumer Loans (% TA)		
Δ Borr. Res. (% TA)	0.00932**	0.0106**	0.0106**	0.00668**
	(0.00362)	(0.00442)	(0.00459)	(0.00304)
Δ Borr. Res. (% TA)$_{t-1}$			0.000892	
			(0.00448)	
Observations	1,774	1,715	1,689	1,715
R-squared	0.369	0.516	0.520	0.517
Unit FE	Bank	Bank	Bank	Bank
Time FE	Quarter	Quarter-Country	Quarter-Country	Quarter-Country
Total assets	Lagged	Lagged	Lagged	Current

Notes: Standard errors clustered at the bank level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

increases in central bank funding are associated with comparatively modest contemporaneous expansions in lending.

6 Firm-Level Analysis

I next investigate how changes in central bank funding are associated with the financial and investment decisions of firms linked to the borrowing banks. Using the matched firm–bank dataset described in Section 4, I examine whether firms connected to banks that increase their borrowed reserves experience larger changes in debt, deposits, and investment. The analysis covers the period 2020–2023, for which annual changes in both bank-level and firm-level balance sheet variables can be constructed. This analysis allows me to extend the previous bank-level results by examining whether additional central bank funding is associated not only with greater firm borrowing, but also with changes in firms' liquidity holdings and real investment.

I estimate the following two-way fixed effects specification:

$$\Delta Y(\%TA)_{i,t} = \beta D_{i,t} + FE_i + FE_t + \epsilon_{i,t},$$

where $Y(\%TA)_{i,t}$ denotes either debt, deposits, or fixed assets of firm i in year t, and $D_{i,t}$

measures the firm's exposure to changes in central bank funding through its linked banks. The dependent variables are expressed as annual changes normalized by lagged total assets, following the normalization introduced in Section 5. The specification includes firm fixed effects, FE_i, which absorb time-invariant differences across firms, and year fixed effects, FE_t, which capture common macroeconomic developments. Standard errors are clustered at the firm level.

I construct firm-level exposure to changes in central bank funding using the firm–bank links described in Section 4.2. For each firm i, I define exposure as the average annual change in borrowed reserves of its linked banks, normalized by their respective lagged total assets:

$$D_{i,t} = \frac{1}{N_i} \sum_{b \in \mathcal{B}_i} \frac{BR_{b,tQ4} - BR_{b,(t-1)Q4}}{TA_{b,(t-1)Q4}} \times 100,$$

where $\mathcal{B}_i$ denotes the time-invariant set of banks linked to firm i for which the necessary balance-sheet information is available, N_i is the number of banks in this set, and $BR_{b,tQ4}$ and $TA_{b,tQ4}$ denote the borrowed reserves and total assets of bank b in the fourth quarter of year t, respectively.

I assign equal weight to all observed banking relationships because the Orbis data report the existence of firm–bank links but not the amount borrowed from each bank. Furthermore, the distribution of a firm's existing borrowing across its banks need not reflect how additional credit is allocated following a change in banks' funding conditions. The unweighted average therefore provides a transparent measure of exposure without imposing assumptions about the relative importance of individual banking relationships.

I estimate five alternative specifications. The first includes firm and year fixed effects, while the second, my preferred specification, replaces year fixed effects with country-by-year fixed effects to account for country-specific macroeconomic conditions and changes in credit demand. The third introduces country-by-sector-by-year fixed effects to control for shocks affecting firms operating in the same sector and country in a given year. The fourth additionally includes lagged exposure to changes in borrowed reserves to examine whether firms' financial and investment decisions respond with a delay, while retaining country-by-year fixed effects.

Finally, in column (5), I construct an alternative measure of firm-level exposure by replacing changes in borrowed reserves with changes in loans to non-financial corporations in the balance sheets of linked banks. Specifically, I compute the unweighted average of the annual changes in NFC loans across each firm's linked banks, normalized by their respective lagged total assets. While the baseline measure captures changes in central bank funding obtained by the banks, this alternative measure captures changes in their lending activity. This allows me to examine whether firms connected to banks expanding their corporate loan portfolios also experience larger changes in their own debt, deposits, and investment.

Table 4 reports the results for firm debt growth. Column (1) shows a positive and statistically significant association between changes in borrowed reserves of linked banks and firms' indebtedness, with an estimated coefficient of 0.0643. The coefficient decreases to 0.0373 when country-by-year fixed effects are introduced in column (2), my preferred specification, but remains statistically significant at the 1 percent level. This estimate implies that a one-percentage-point increase in the average borrowed reserves of a firm's linked banks, relative to their total assets, is associated with an increase in firm debt of 0.0373 percentage points of its total assets. The coefficient remains positive and statistically significant when introducing country-by-sector-by-year fixed effects in column (3). In column (4), the contemporaneous coefficient decreases to 0.0151 and is significant at the 10 percent level, while the lagged coefficient is not statistically significant.

Finally, column (5) examines whether firms connected to banks expanding their lending to non-financial corporations also experience greater debt growth. The estimated coefficient is positive and statistically significant at the 1 percent level (0.0588), indicating that a one-percentage-point increase in the average NFC lending of a firm's linked banks, relative to their total assets, is associated with an increase in firm debt of approximately 0.059 percentage points of total assets. This finding confirms a positive relationship between changes in banks' corporate loan portfolios and the indebtedness of their linked firms.

Table 5 reports the results for firm deposit growth. The estimated coefficients are positive and statistically significant across all specifications. In column (1), the coefficient is 0.0169 and increases to 0.0268 when country-by-year fixed effects are introduced in column (2), my preferred specification. This estimate indicates that a one-percentage-point

Table 4: Borrowed Reserves and Firm Debt Growth

	(1)	(2)	(3)	(4)	(5)
Dependent variable			Δ Debt (% TA)		
Δ Borr. Res. (% TA)	0.0643***	0.0373***	0.0327***	0.0151*	
	(0.00631)	(0.00701)	(0.00700)	(0.00816)	
Δ Borr. Res. (% TA)$_{t-1}$				-0.00509	
				(0.00759)	
Δ NFC Loans (% TA)					0.0588***
					(0.00910)
Observations	762,609	762,609	762,052	517,803	762,609
R-squared	0.329	0.334	0.339	0.387	0.334
Unit FE	Firm	Firm	Firm	Firm	Firm
Time FE	Year	Year-Country	Year-Country-Sector	Year-Country	Year-Country

Notes: Standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

increase in the average borrowed reserves of a firm's linked banks, relative to their total assets, is associated with an increase in firm deposits of approximately 0.027 percentage points of total assets. The coefficient remains positive and statistically significant when introducing country-by-sector-by-year fixed effects in column (3). In column (4), the contemporaneous coefficient is similar to that of the preferred specification, while the lagged exposure coefficient is also positive and weakly significant, suggesting that the association between central bank funding and firms' deposit accumulation may extend beyond the contemporaneous year.

Finally, column (5) investigates whether firms connected to banks expanding their corporate lending also accumulate more deposits. The estimated coefficient is positive and statistically significant at the 1 percent level (0.0503), indicating that a one-percentage-point increase in the average NFC lending of a firm's linked banks, relative to their total assets, is associated with an increase in firm deposits of approximately 0.050 percentage points of total assets. This suggests that firms connected to banks expanding their corporate loan portfolios experience not only greater indebtedness, but also greater liquidity accumulation.

Table 6 reports the results for firm investment, proxied by changes in fixed assets. Column (1) shows a negative and statistically significant coefficient of -0.0131. However,

Table 5: Borrowed Reserves and Firm Deposits Growth

	(1)	(2)	(3)	(4)	(5)
Dependent variable			Δ Deposits (% TA)		
Δ Borr. Res. (% TA)	0.0169**	0.0268***	0.0215***	0.0272***	
	(0.00726)	(0.00745)	(0.00746)	(0.00985)	
Δ Borr. Res. (% TA)$_{t-1}$				0.0158*	
				(0.00892)	
Δ NFC Loans (% TA)					0.0503***
					(0.00970)
Observations	1,053,127	1,053,127	1,052,775	692,108	1,053,127
R-squared	0.241	0.242	0.248	0.296	0.242
Unit FE	Firm	Firm	Firm	Firm	Firm
Time FE	Year	Year-Country	Year-Country-Sector	Year-Country	Year-Country

Notes: Standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

the coefficient becomes positive and statistically insignificant when country-by-year fixed effects are introduced in column (2), my preferred specification. The estimated coefficient of 0.00381 indicates that there is no statistically significant association between changes in borrowed reserves of linked banks and firm investment once country-specific macroeconomic conditions are accounted for. This finding remains robust to the inclusion of country-by-sector-by-year fixed effects in column (3). In column (4), neither contemporaneous nor lagged exposure is statistically significant, providing no evidence of an investment response in the current or subsequent year. Finally, column (5) examines whether firms connected to banks expanding their lending to non-financial corporations experience greater investment. The estimated coefficient is close to zero (0.00140) and statistically insignificant, suggesting that even increases in banks' corporate lending are not associated with higher investment among their linked firms.

Taken together, these results reveal a clear distinction between firms' financial and real responses to changes in their lending banks' central bank borrowing. Firms connected to banks that increase their borrowed reserves experience significantly higher debt and deposit growth, but no corresponding increase in investment. This pattern is consistent with firms retaining at least part of the additional financing as liquidity rather than immediately deploying it towards productive investment. Importantly, the absence of a significant investment response also holds when firm-level exposure is constructed using

Table 6: Borrowed Reserves and Firm Investment

	(1)	(2)	(3)	(4)	(5)
Dependent variable			Δ Fixed Assets (% TA)		
Δ Borr. Res. (% TA)	-0.0131**	0.00381	0.00376	0.00126	
	(0.00562)	(0.00589)	(0.00591)	(0.00763)	
Δ Borr. Res. (% TA)$_{t-1}$				-0.00459	
				(0.00714)	
Δ NFC Loans (% TA)					0.00140
					(0.00785)
Observations	1,205,172	1,205,172	1,204,633	796,686	1,205,172
R-squared	0.334	0.334	0.335	0.398	0.334
Unit FE	Firm	Firm	Firm	Firm	Firm
Time FE	Year	Year-Country	Year-Country-Sector	Year-Country	Year-Country

Notes: Standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

changes in banks' lending to non-financial corporations, rather than their central bank borrowing.

7 Event Study

7.1 Treatment Definition and Empirical Specification

I now turn to the pandemic recalibration of TLTRO III to investigate the effects of a specific expansion in central bank funding on firms' financial and investment decisions. As discussed in Section 3, beginning in March 2020, the ECB substantially relaxed the programme's borrowing conditions, triggering a sharp increase in central bank funding across the euro area. However, banks differed considerably in the extent to which they increased their borrowing from the Eurosystem. I exploit this cross-sectional variation to implement a continuous-treatment difference-in-differences design, comparing the evolution of firms' debt, deposits, and investment before and after the policy intervention across firms with different degrees of exposure.

Unlike the previous section, where firm-level exposure varies annually, I now construct a time-invariant measure of exposure to the pandemic recalibration. Specifically, I measure each bank's treatment intensity as the increase in its borrowed reserves between 2019Q4

and 2020Q4, normalized by its total assets at the end of 2019. I then construct firm-level exposure, denoted by D_i, as the unweighted average of the treatment intensities of its linked banks:

$$D_i = \frac{1}{N_i} \sum_{b \in \mathcal{B}_i} \frac{BR_{b,2020Q4} - BR_{b,2019Q4}}{TA_{b,2019Q4}} \times 100,$$

where $\mathcal{B}_i$ denotes the time-invariant set of banks linked to firm i for which the necessary balance-sheet information is available, N_i is the number of banks in this set, and $BR_{b,t}$ and $TA_{b,t}$ denote borrowed reserves and total assets of bank b at date t, respectively.

I focus on the change between 2019Q4 and 2020Q4 because this window captures the sharp expansion in central bank funding associated with the pandemic recalibration of TLTRO III. Using the change in borrowed reserves, rather than their level, allows me to abstract from persistent differences across banks in their pre-existing reliance on central bank funding. Scaling this change by total assets measured at the end of 2019 makes the treatment intensity comparable across banks of different sizes and ensures that the denominator is predetermined with respect to the policy intervention.

I then estimate the following continuous-treatment difference-in-differences specification:

$$\Delta Y(\%TA)_{i,t} = \sum_{s=2016}^{2018} \phi_s \left(D_i \times \mathbb{1}\{t = s\}\right) + \sum_{s=2020}^{2023} \tau_s \left(D_i \times \mathbb{1}\{t = s\}\right) + FE_i + FE_{c,t} + \epsilon_{i,t}, \tag{1}$$

where $\Delta Y(\%TA)_{i,t}$ denotes the annual change in firm debt, deposits, or fixed assets, normalized by lagged total assets as defined in Section 5. The specification includes firm fixed effects, FE_i, which absorb time-invariant differences across firms, and country-by-year fixed effects, $FE_{c,t}$, which account for country-specific macroeconomic conditions and other shocks common to firms within the same country and year. The omitted reference year is 2019. The coefficients ϕ_s capture pre-treatment differences in outcome changes associated with future exposure, while τ_s capture the corresponding differences following the pandemic recalibration. Standard errors are clustered at the firm level.

7.2 Identification Assumptions

The causal interpretation of the estimates relies on the identifying assumptions for difference-in-differences designs with a continuous treatment discussed by Wooldridge (2023) and Callaway et al. (forthcoming).

First, I assume that firms are independently and identically sampled from the underlying population. This assumption is particularly relevant in my setting because the final estimation sample results from several data-availability and matching restrictions. Firms may be excluded because firm–bank links, bank-level information, or outcome variables are not observed. The identifying assumption requires that this selection does not induce systematic differences in the joint distribution of treatment exposure and potential outcomes between firms included in and excluded from the estimation sample.

Second, I impose a no-anticipation assumption. Let $Y_{i,t}(d)$ denote the potential outcome of firm i in year t as a function of treatment intensity d. For every pre-treatment year $t < 2020$, firms' potential outcomes are assumed to be invariant to the treatment intensity they subsequently receive, so that

$$Y_{i,t}(d) = Y_{i,t}(0) \tag{2}$$

for all d in the support of D_i. In other words, firms that are subsequently more exposed to TLTRO III should not adjust their financing or investment decisions before the pandemic recalibration in anticipation of their future treatment intensity. This assumption is particularly plausible in this setting because the relevant recalibration of TLTRO III was introduced in response to the COVID-19 shock, making the subsequent increase in the attractiveness and availability of central bank funding difficult to anticipate before 2020.

Third, I impose a strong parallel trends assumption appropriate for the continuous-treatment setting. Following Wooldridge (2023), I assume that the potential outcome $Y_{i,t}(d)$ is linear in treatment intensity d, and let $B_{i,t}$ denote the slope of this relationship. The strong parallel trends assumption requires the expected treatment-effect slope to be independent of treatment intensity,

$$E[B_{i,t} \mid D_i] = E[B_{i,t}]. \tag{3}$$

This allows treatment effects to be heterogeneous across firms, but rules out systematic selection into different treatment intensities on the basis of the per-unit gains from treatment. In my setting, this assumption requires that firms connected to banks with higher TLTRO III exposure are not systematically those for which an additional unit of exposure has a larger or smaller effect on debt, deposits, or investment. Firms may differ in their sensitivity to bank funding, and banks may differ in their use of TLTRO III, but the intensity of treatment cannot be systematically related to these heterogeneous per-unit gains.

Under these assumptions, the post-treatment coefficient τ_s identifies $E[B_{i,s}]$, the average slope of the potential-outcome function in year s with respect to treatment intensity. The coefficient τ_s can therefore be interpreted as the average change in a firm's outcome $Y_{i,s}$ in year s associated with all banks linked to that firm increasing their central bank borrowing by one percentage point of their total assets.

While the identifying assumptions cannot be tested directly, the pre-treatment coefficients ϕ_s provide evidence on their plausibility. In particular, coefficients close to zero before 2020 indicate that firms with different subsequent treatment intensities did not exhibit systematically different trends prior to the pandemic recalibration.

7.3 Results

This section presents the results of the continuous-treatment difference-in-differences analysis, examining how firms' debt, deposits, and investment respond to the pandemic recalibration of TLTRO III. Table 7 reports the estimated coefficients, while Figures 6, 7, and 8 illustrate their evolution over time.

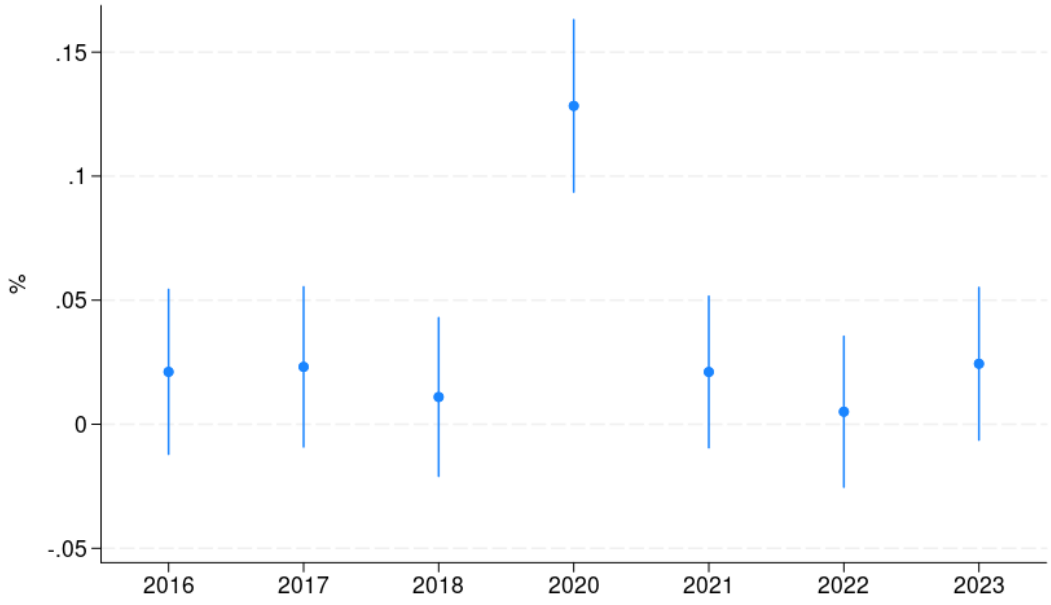
I first examine the response of firm debt. Figure 6 and column (1) of Table 7 show that the pre-treatment coefficients are small and statistically insignificant, providing no evidence of differential trends in debt growth between firms with different levels of subsequent exposure to TLTRO III. In 2020, however, the estimated coefficient increases sharply to 0.128 and is statistically significant at the 1 percent level. This indicates that firms

Table 7: The Effects of the Pandemic Recalibration of TLTRO III

	(1) Δ Debt (% TA)	(2) Δ Deposits (% TA)	(3) Δ Fixed Assets (% TA)
$\hat{\phi}_{2016}$	0.0211	0.0223	-0.00892
	(0.0169)	(0.0145)	(0.0123)
$\hat{\phi}_{2017}$	0.0231	0.0200	0.00567
	(0.0164)	(0.0144)	(0.0120)
$\hat{\phi}_{2018}$	0.0110	0.0100	-0.00816
	(0.0163)	(0.0151)	(0.0115)
$\hat{\tau}_{2020}$	0.128***	0.0519***	0.00138
	(0.0177)	(0.0163)	(0.0111)
$\hat{\tau}_{2021}$	0.0211	0.000495	-0.0169
	(0.0156)	(0.0150)	(0.0115)
$\hat{\tau}_{2022}$	0.00505	-0.0114	-0.0144
	(0.0155)	(0.0148)	(0.0116)
$\hat{\tau}_{2023}$	0.0244	-0.0495***	-0.0214*
	(0.0157)	(0.0148)	(0.0118)
Observations	1,778,598	2,581,814	2,922,442
R-squared	0.211	0.125	0.202
Unit FE	Firm	Firm	Firm
Time FE	Country-Year	Country-Year	Country-Year

Notes: The table reports estimates from the continuous-treatment difference-in-differences specification in Equation 1. Firm-level exposure is constructed from the change in borrowed reserves of linked banks between 2019Q4 and 2020Q4. Dependent variables are annual changes normalized by lagged total assets and winsorized at the 1st and 99th percentiles. The omitted reference year is 2019. Standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 6: Dynamic Effect of Pandemic TLTRO III Exposure on Firm Debt Growth



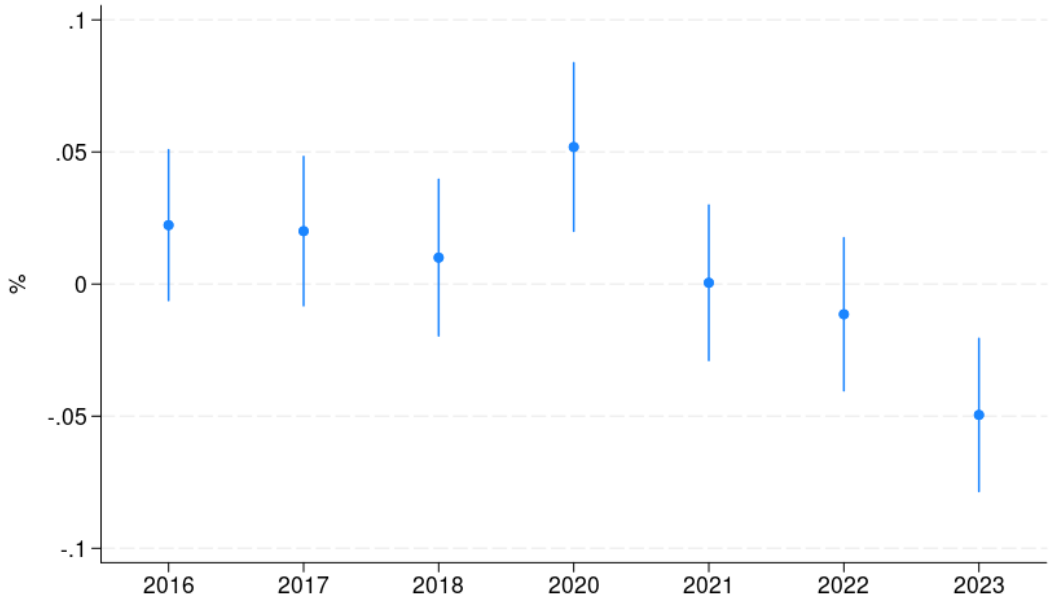
Notes: The figure reports the estimated dynamic coefficients from the baseline continuous-treatment difference-in-differences specification for firm debt growth. Firm-level exposure is defined as the average exposure to the pandemic recalibration of TLTRO III among the banks linked to each firm. The dependent variable is the annual change in firm debt scaled by lagged total assets. The omitted reference year is 2019. Vertical bars report 95 percent confidence intervals based on standard errors clustered at the firm level.

more exposed to banks increasing their central bank borrowing experienced significantly higher debt growth following the pandemic recalibration. The coefficients for 2021, 2022, and 2023 are substantially smaller and statistically insignificant, suggesting that the increase in firm borrowing was concentrated in 2020, with little evidence of additional debt accumulation in subsequent years.

Quantitatively, the 2020 coefficient implies that a one-percentage-point increase in TLTRO III exposure is associated with an increase in firm debt of 0.128 percentage points of total assets. For a firm with median exposure of 5.28 and approximately €1.2 million in total assets, this corresponds to an increase in debt of approximately €8,100 in 2020.

Turning to deposits, column (2) of Table 7 and Figure 7 show a similar initial response. The pre-treatment coefficients are small and statistically insignificant, providing no evidence of differential trends in deposit growth before the intervention. In 2020, the coeffi-

Figure 7: Dynamic Effect of Pandemic TLTRO III Exposure on Firm Deposit Growth

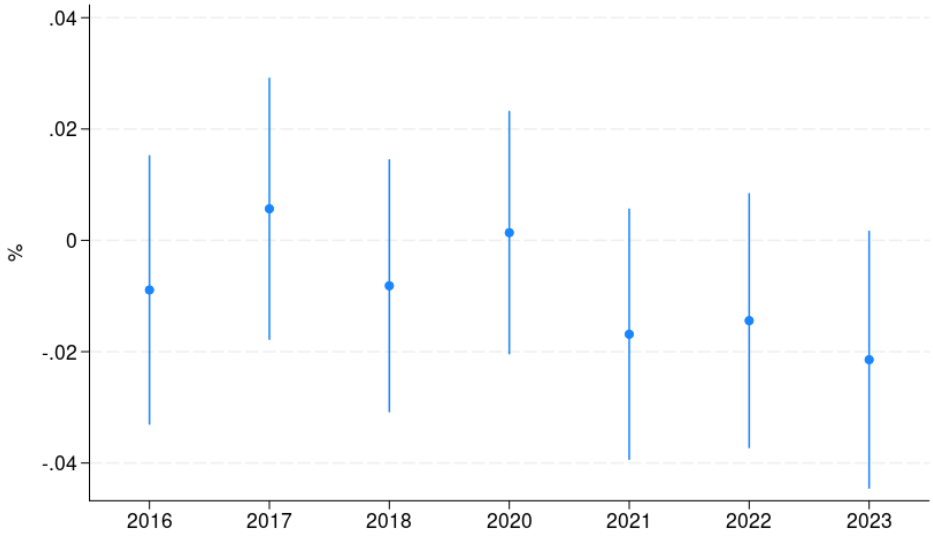


Notes: The figure reports the estimated dynamic coefficients from the baseline continuous-treatment difference-in-differences specification for firm deposit growth. Firm-level exposure is defined as the average exposure to the pandemic recalibration of TLTRO III among the banks linked to each firm. The dependent variable is the annual change in firm deposits scaled by lagged total assets. The omitted reference year is 2019. Vertical bars report 95 percent confidence intervals based on standard errors clustered at the firm level.

cient is positive and statistically significant at the 1 percent level (0.0519), indicating that firms more exposed to the pandemic recalibration also experienced higher deposit growth. The coefficients for 2021 and 2022 are close to zero and statistically insignificant. In 2023, however, the coefficient becomes negative and statistically significant at the 1 percent level (-0.0495), indicating that more exposed firms experienced lower deposit growth relative to less exposed firms.

For a firm with median exposure and approximately €1.2 million in total assets, the estimated coefficients imply an increase in deposits of around €3,300 in 2020 and a relative reduction in deposit growth of approximately €3,100 in 2023. These results suggest that firms initially accumulated part of the additional financing as liquidity, although the negative coefficient in 2023 indicates that this pattern did not persist throughout the post-treatment period.

Figure 8: Dynamic Effect of Pandemic TLTRO III Exposure on Firm Investment



Notes: The figure reports the estimated dynamic coefficients from the baseline continuous-treatment difference-in-differences specification for firm investment. Firm-level exposure is defined as the average exposure to the pandemic recalibration of TLTRO III among the banks linked to each firm. The dependent variable is the annual change in fixed assets scaled by lagged total assets and is used as a proxy for investment. The omitted reference year is 2019. Vertical bars report 95 percent confidence intervals based on standard errors clustered at the firm level.

Finally, column (3) of Table 7 and Figure 8 report the response of firm investment, proxied by changes in fixed assets. The pre-treatment coefficients are small and statistically insignificant, providing no evidence of differential trends in investment prior to the pandemic recalibration. Unlike debt and deposits, however, investment does not exhibit a positive and statistically significant response in 2020. The estimated coefficient is close to zero (0.00138) and statistically insignificant. The coefficients remain negative and statistically insignificant in 2021 and 2022, while the coefficient for 2023 is negative (-0.0214) and significant at the 10 percent level.

These findings provide no evidence that the additional financing associated with the pandemic recalibration of TLTRO III translated into higher firm investment. If anything, the negative coefficient in 2023 suggests a possible reduction in investment among more exposed firms, although this result is only weakly statistically significant.

Taken together, the event-study results reveal a clear distinction between the financial and

real effects of the pandemic recalibration of TLTRO III. Firms connected to banks that increased their central bank borrowing experienced significantly higher debt and deposit growth in 2020, but no corresponding increase in investment. While the initial expansion in borrowing is consistent with the intended transmission of targeted central bank funding to firms, the absence of a positive investment response suggests that the programme did not necessarily stimulate productive capital accumulation. These findings are consistent with the results of the previous two-way fixed effects analyses and highlight the importance of looking beyond bank lending when evaluating the effectiveness of funding-for-lending programmes.

8 Conclusion

This paper examines the real effects of funding-for-lending programmes by investigating how central bank funding is transmitted through banks to firms and whether it ultimately translates into greater investment. Using the ECB's Targeted Longer-Term Refinancing Operations (TLTRO III) and a novel matched firm–bank dataset covering 470,000 firms across 14 euro-area countries, I trace the relationship between banks' central bank borrowing, bank lending, and firms' financial and investment decisions.

The results reveal a clear distinction between the financial and real effects of targeted central bank funding. Banks that increase their borrowing from the Eurosystem expand their lending, particularly to non-financial corporations. Firms connected to these banks experience significantly higher debt and deposit growth, but no corresponding increase in investment. The event-study analysis exploiting the pandemic recalibration of TLTRO III confirms this pattern: more exposed firms increase their borrowing and accumulate liquidity following the intervention, while investment does not respond positively.

These findings highlight an important limitation of funding-for-lending programmes. Although improving banks' funding conditions is associated with an expansion in credit, this does not necessarily translate into greater productive investment. Evaluating the effectiveness of such programmes therefore requires looking beyond bank lending and examining how firms ultimately use the additional financing.

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